

Alexandre EL ASHI

Mechanical Engineering - EPFL

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Eager to build a career in Mechanical Engineering, I bring a unique perspective shaped by a strong technical background. My passion for artificial intelligence, machine learning and optimization is complemented by innovative thinking, **multilingual** capabilities, solid **managerial skills**, and proficiency in hands-on laboratory practices. Open-minded and **adaptable**, I am committed to learning and contributing effectively within a team to excel in any dynamic and innovative industry.

Education

- **ETHZ - Zurich, Switzerland** 2026 - 2028
 - Master’s degree in Mechanical Engineering specialized in AI and optimization.
 - **EPFL - Lausanne, Switzerland** 2023 - 2026
 - Bachelor’s degree in Mechanical Engineering (grade 5.00/6.00 Bachelor Cycle)

Professional Experience

- **MYSTONES GROUP Sàrl – Lausanne, Switzerland** (Summer 2026)
 - Developed a Python-based computer vision pipeline using Sports2D and deep neural network-based pose estimation to extract joint keypoints and biomechanical metrics from athlete videos.
 - Implemented statistical sensitivity methods (linear regression, MDC thresholds and z-scores) to detect meaningful performance deviations and integrated the automated pipeline into the platform’s backend.
 - **The Consulting Society, EPFL - Lausanne, Switzerland** (2024 - 2026)
 - At DualEnergy (agrivoltaics start-up), I conducted market and feasibility studies covering crop compatibility, land suitability, and regulatory constraints for solar deployment. I also developed MATLAB models to compute Weibull probability distributions for solar panel failure analysis.
 - **AMAC, EPFL - Lausanne, Switzerland** (2025 - 2026)
 - Treasurer of the association of mechanical engineer at EPFL (Accounting + Excel)
 - **MEDITEX Logistix SAL - Beirut, Lebanon** (Summer 2025)
 - Suggested and implemented RFID and GPS tracking systems to optimize hospital linen logistics, improve efficiency, and support financial oversight.
 - **Bemo Securisation SAL BSEC - Beirut, Lebanon** (Summer 2022)
 - Completed an internship in investment banking, where I discovered the world of finance and actively contributed to several multi-million projects. Gained hands-on experience with key financial indicators such as CAPEX, EBITDA, and others, developing a solid understanding of financial analysis and strategic investment processes.

Projects

- **Raspberry Sorting Robot, EPFL** (Fall 2025)
 - Designed and implemented a robotic system for raspberry sorting using computer vision for color-based detection and size estimation.
 - **Graphical Data Rectification, EPFL** (Fall 2024)
 - Worked on a multidisciplinary project focused on image analysis using mathematical tools. The project involved developing and optimizing algorithms in C, MATLAB, and LabVIEW to process and analyze graphic images.

Extra-curricular Activities

- **Coach and Mentor - Coaching EPFL** 2024 - 2026
 - Mentor and support first-year students at EPFL, helping them integrate into campus life both academically and socially. Provide guidance, answer questions, and organize activities to foster a welcoming and inclusive environment.
 - **Class representative - EPFL** 2023 - 2024
 - Act as the liaison between students and professors, representing the interests and concerns of the class. Facilitate communication, provide feedback, and help maintain a productive academic environment.
 - **Athlete**
 - **Professional Skier:** Part of the Lebanese National Selection U16, played in various championship. 2009 - 2019
 - **Triathlete:** Currently training for an IRONMAN 70.3 Ongoing
 - **Marathoner:** Lausanne Marathon Ongoing

Personnal Info and Languages

- **French** (Fluent)
 - **English** (Fluent)
 - **Arabic** (Native)
 - Technical skills: Matlab, LabView, C, CAD, Overleaf, Python, Excel, Abaqus.